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Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 18

Name of Student	Abobakr Mohamed elameen
PRN No.	25070122015
Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year –3rd Sem
Title of Assignment	Write a Java GUI program to demonstrate button click event handling.
Practice questions	1. Create a GUI calculator where buttons perform addition and subtraction.

```java
import javax.swing.*;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;

public class CalculatorDemo {
    public static void main(String[] args) {
        JFrame frame = new JFrame("GUI Calculator");
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        JTextField num1Field = new JTextField();
        num1Field.setBounds(30, 20, 100, 25);
        frame.add(num1Field);

        JTextField num2Field = new JTextField();
        num2Field.setBounds(150, 20, 100, 25);
        frame.add(num2Field);

        JButton addButton = new JButton("Add (+)");
        addButton.setBounds(30, 60, 100, 25);
        frame.add(addButton);
```

```
        JButton subButton = new JButton("Subtract (-)");
        subButton.setBounds(150, 60, 100, 25);
        frame.add(subButton);

        JLabel resultLabel = new JLabel("Result: ");
        resultLabel.setBounds(30, 110, 220, 25);
        frame.add(resultLabel);

        addButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double a = Double.parseDouble(num1Field.getText());
                double b = Double.parseDouble(num2Field.getText());
                resultLabel.setText("Result: " + (a + b));
            }
        });

        subButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double a = Double.parseDouble(num1Field.getText());
                double b = Double.parseDouble(num2Field.getText());
                resultLabel.setText("Result: " + (a - b));
            }
        });

        frame.setVisible(true);
    }
}
```

2. Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.

```
import javax.swing.*;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;

public class BankBalanceCalculator {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Bank Balance Calculator");
        frame.setSize(320, 220);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        JLabel balanceLabel = new JLabel("Initial Balance:");
        balanceLabel.setBounds(20, 20, 120, 25);
        frame.add(balanceLabel);
```

```
        JTextField balanceField = new JTextField();
        balanceField.setBounds(150, 20, 130, 25);
        frame.add(balanceField);

        JLabel amountLabel = new JLabel("Transaction Amount:");
        amountLabel.setBounds(20, 60, 120, 25);
        frame.add(amountLabel);

        JTextField amountField = new JTextField();
        amountField.setBounds(150, 60, 130, 25);
        frame.add(amountField);

        JButton depositButton = new JButton("Deposit");
        depositButton.setBounds(30, 100, 100, 25);
        frame.add(depositButton);

        JButton withdrawButton = new JButton("Withdraw");
        withdrawButton.setBounds(150, 100, 100, 25);
        frame.add(withdrawButton);

        JLabel resultLabel = new JLabel("Updated Balance: ");
        resultLabel.setBounds(20, 150, 250, 25);
        frame.add(resultLabel);

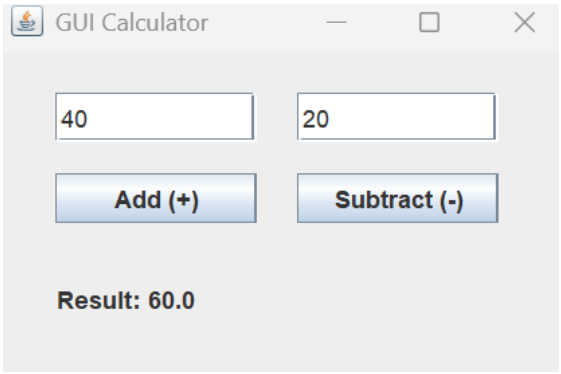
        depositButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double balance =
Double.parseDouble(balanceField.getText());
                double amount =
Double.parseDouble(amountField.getText());
                double updatedBalance = balance + amount;
                resultLabel.setText("Updated Balance: " +
updatedBalance);
                balanceField.setText(String.valueOf(updatedBalance))
;
            }
        });

        withdrawButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                double balance =
Double.parseDouble(balanceField.getText());
                double amount =
Double.parseDouble(amountField.getText());
                double updatedBalance = balance - amount;
                resultLabel.setText("Updated Balance: " +
updatedBalance);
                balanceField.setText(String.valueOf(updatedBalance))
;
            }
        });
```

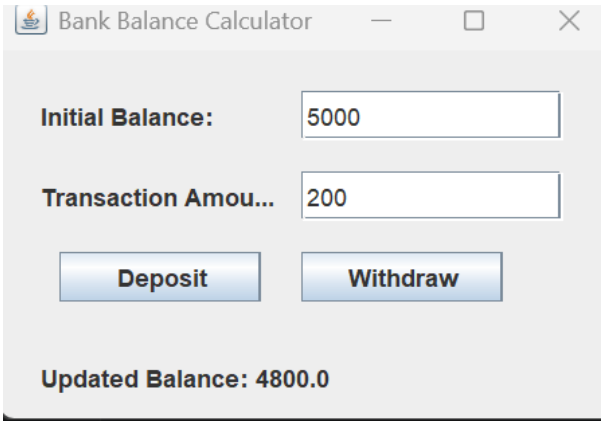
```
        frame.setVisible(true);
    }
}
```

Output

1) After Clicking 'Add'



2) After Clicking 'Withdraw'



Conclusion

Event handling is what makes a GUI interactive — without it, buttons would just sit there. Using `ActionListener` and its `actionPerformed()` method, a Java Swing application can respond immediately to user actions like button clicks, separating the interface layout from the logic that processes user input, as shown in the calculator and bank balance examples.